

Dose-Finding: 3 cultures × 2 temperatures

Three independently started cultures, each split into two identical plates, one at 15 °C and one at 27 °C. No drug. Six plates answer two questions at once: the working dose of each inhibitor, and whether the inhibitors cost growth only in the heat (the AOX-engagement signature). Culture = biological replicate; never pool or share diluted cells between cultures.

3 cultures × 2 plates = 6 plates

DF_15_1 / _2 / _3 · 15 °C

DF_27_1 / _2 / _3 · 27 °C

no prothioconazole in cell wells

15 wells per plate · 90 vials

GATE before anything goes in a vial

- 1. **Each culture OD ≥ 0.10** and healthy under the scope. Log OD and age per culture. A culture below 0.10 waits; the others can run.
- 2. **Reader calibrated** (two-point) on the washed vials. One vial of plain air-shaken YMS must read ~7-8 mg/L. If not, fix that first.
- 3. nPG master made **fresh today**; SHAM master from the foil-wrapped stock. Both clear.

PAIRS one culture → one pair of plates

CULTURE	15 °C PLATE	27 °C PLATE	CELLS TO MAKE
culture 1	DF_15_1	DF_27_1	~130 mL @ OD 0.0005
culture 2	DF_15_2	DF_27_2	~130 mL @ OD 0.0005
culture 3	DF_15_3	DF_27_3	~130 mL @ OD 0.0005

MAP identical layout on all six plates

DF_15_n
15 °C incubator

DF_27_n
27 °C incubator

	1	2	3	4	5	6
A	A1 CTRL EtOH	A2 CTRL EtOH	A3 nPG 0.0125 mM	A4 nPG 0.025 mM	A5 nPG 0.05 mM	A6 nPG 0.1 mM
B	B1 SHAM 0.0125 mM	B2 SHAM 0.025 mM	B3 SHAM 0.05 mM	B4 SHAM 0.1 mM	B5 SHAM 0.2 mM	B6



■ control (cells + ethanol)
 ■ nPG ladder
 ■ SHAM ladder
 ■ cell-free blanks (C1-C4)

C3-C4 show the drug itself doesn't move the sensor chemistry, so blank-subtraction in the main run is defensible. 11 cell wells + 4 blanks = 15 per plate, 90 total.

FILL every well the same, one thing changes

cell wells (A1-B5): 5.03 mL cells (OD 0.0005) + 20 µL P0 (DMSO) + 10 µL inhibitor stock (or plain EtOH for A1-A2)

blanks C1-C2: 5.03 mL sterile YMS + 20 µL P0 + 10 µL inhibitor

blanks C3-C4: 5.03 mL sterile YMS + 20 µL P4 (1 mg/mL drug) + 10 µL EtOH (C3) / 10 µL 50 mM nPG (C4)

Cells, per culture: **~130 mL at OD 0.0005** (mL culture = $0.0005 \times 130 \div$ that culture's OD, top up with YMS). Fill that culture's two plates alternately from its own batch, swirling between draws. Three separate batches, three separate labelled bottles. Foil everything with nPG/SHAM. Cap, invert gently, no bubbles.

STOCKS ethanol dilutions of your masters — 10 µL/well, one set serves all six plates (~1 mL each)

FINAL	NPG STOCK (FROM 50 MM)	SHAM STOCK (FROM 250 MM)
0.0125 mM	6.25 mM · 125 µL+875 EtOH	6.25 mM · 25 µL+975 EtOH
0.025 mM	12.5 mM · 250 µL+750 EtOH	12.5 mM · 50 µL+950 EtOH
0.05 mM	25 mM · 500 µL+500 EtOH	25 mM · 100 µL+900 EtOH
0.1 mM	50 mM · master neat	50 mM · 200 µL+800 EtOH
0.2 mM	— (nPG stops at 0.1)	100 mM · 400 µL+600 EtOH

Not serial: every stock is made straight from its master. All doses sit 2.5-40× below the 0.5 mM SHAM that abolished growth last run.

RUN

1. Work culture by culture: gate → dilute → fill its pair → into the readers. Then the next culture. All six started the same afternoon. Dark.
2. Export names: **DF_15_1..3_Oxygen**, **DF_27_1..3_Oxygen**. Log per culture: OD, age, start time; plus the calibration reading.
3. 27 °C turns over in ~4 days; 15 °C in ~5-6. Harvest by the curves, not the clock.
4. Same day, start the **induced flask** (27 °C + 2 mg/L drug, ~50 mL, inoculate ~OD 0.02) for the azide plate.

READ two answers from one pair of plates

The dose (27 °C plate)

Growth vs inhibitor dose. Working dose = highest well within ~15-20% of A1-A2, confirmed later by the azide plate as actually blocking. Carry **two** doses into the main run.

The split (both plates)

Same ladder, two temperatures. Inhibitor costing growth at 27 °C but not at 15 °C = AOX engaged only in the heat. Replicated on three days, this is a figure on its own.

Three cultures on one day share medium batch, stocks and reader session, so day and culture are confounded. Legitimate biological replicates all the same; one extra culture in a later week makes it bulletproof if a reviewer asks.